

pdfexplore TO-DO

Target

pdfexplore should feel like the PDF analog of mdexplore : same root-navigation shell, same scope rules where PDFs make sense, same sidecar philosophy, and the same habit of preserving work state without modifying source documents.

Current Snapshot

- Search and prefetch now run on worker pools with cancellation support.
- Text extraction uses layered cache (memory + disk) and badge visibility.
- Tree marker state merges sidecar scan + live state to reduce stale badge drops.
- Interaction-first throttling is in place (scroll and tree mutation pauses).
- Viewer right-rail search markers now build progressively with document-scoped page-text caching.
- Marker builds now include interaction-priority behavior so click-to-jump remains responsive during long scans.

Roadmap (2026)

This roadmap is organized by execution horizon so contributors can prioritize work without reinterpreting old phase notes.

Now (next 1-3 weeks)

- ☐ Finalize PDF top-button behavior and confirm UX contract against mdexplore parity expectations.
- ☐ Run manual UX pass for search-hit/effective-root pill styling and capture any visual regressions.
- ☐ Run manual UX pass for highlight visibility, marker badge continuity, scroll return, and zoom-feel parity.
- ☐ Add focused regression tests for marker badge continuity under file switch and search rerun scenarios.

Next (next 1-2 months)

- ☐ Add performance regression coverage for root changes in medium/large trees.
- ☐ Add deterministic tests for interaction-heavy scenarios where prefetch, marker scan, and search overlap.
- ☐ Introduce optional timing instrumentation for hot paths: marker sync, visible-tree listing, and search aggregation.
- ☐ Refresh top-row control tests after `PDF` behavior is finalized.

Later (quarterly hardening)

- ☐ Evaluate more helper extraction into `mdexplore_app` to reduce divergence and maintenance overhead.
- ☐ Continue multi-instance soak testing for config/recent-roots lock behavior.
- ☐ Reassess cache/prefetch heuristics for very large directory trees and document practical tuning defaults.

Audit Summary

Already working

- PDF-only tree with directory navigation.
- Embedded local `pdf.js` preview.
- File highlight colors with sidecar persistence.
- Preview text highlights with two-tier sidecar persistence using `.pdfexplore-highlighting.json`.
- Multi-view tabs with per-document session restore.
- Single-view return-to-document scroll/session restore.
- Preview zoom in/out/reset with `mdexplore`-matching shortcuts and overlay.
- Copy actions with directory metadata merge.
- Auto-refresh when the currently previewed PDF changes on disk.

Main UX gaps vs mdexplore

1. Shell parity still needs final polish.
 - Recent and Edit are now present.
 - PDF behavior/design parity is still open.
2. Root persistence parity needs continued soak testing.
 - JSON payload and lock strategy are implemented.
 - Recent-root behavior should continue to be validated under multi-instance use.
3. Search/marker performance tuning remains ongoing.
 - Search scope now follows visible-tree behavior.
 - Marker generation throughput and click responsiveness were both improved.
 - Remaining work is mostly calibration/timing for very large PDFs and edge hardware.
4. Visual consistency needs additional regression checks.
 - Badge coexistence (search pill + marker + cache badge) should be regression-tested more thoroughly.
5. Highlight/preview parity still needs manual UX confirmation.
 - Two-tier preview highlight colors use shared palette.
 - Scroll/session return and zoom are implemented; parity feel-check still open.
6. External-file actions are partially settled.
 - Edit opens current document externally.
 - PDF button behavior remains open for long-term parity/design.
7. Test coverage gaps remain for complex UI interactions.
 - Need stronger regressions for marker badge continuity and heavy-interaction throttling.

Detailed Phase Plan

Phase 1

- ☒ Add Recent button and dropdown before ^.
- ☒ Move pdfexplore config to JSON payload with legacy plain-text read support.
- ☒ Add recent-root storage, menu presentation, reload-on-open, dwell timing, cap of 35, and lock-file writes.
- ☒ Add tests for recent-root behavior and config persistence.

Phase 2

- ☒ Rework search to use visible-tree PDF scope instead of one-directory direct children.
- ☒ Rerun active search automatically on expand/collapse/root/scope changes.
- ☐ Manual UX verification for effective-root and search-hit pill styling. Current wiring now uses the shared effective-scope tree model path.

Phase 3

- ☒ Persist preview text highlights in `.pdfexplore-highlighting.json`.
- ☒ Use the shared two-tier preview highlight colors from `mdexplore`.
- ☒ Preserve scroll/session position when returning to a PDF.
- ☒ Implement preview zoom in/out/reset with the `mdexplore` shortcut set.
- ☐ Manual UX verification of highlight visuals, marker badge continuity, scroll return, and zoom feel.

Phase 4

- ☐ Define and implement final `PDF` button behavior for a PDF-native workflow.
- ☒ Implement `Edit` behavior for PDFs.
- ☐ Add/refresh tests for top-row controls once `PDF` behavior is finalized.

Phase 5

- ☐ Review whether more `mdexplore_app` helpers should be shared to reduce divergence between explorers.

Phase 6 (Stability / Responsiveness)

- ☐ Add focused performance regression tests around root changes with medium-size directories.
- ☐ Add deterministic marker-badge tests for transitions: add highlight -> switch file -> switch back.
- ☐ Add viewer-bridge regression tests for marker click responsiveness during in-progress marker builds.
- ☐ Add lightweight optional timing instrumentation hooks for hot paths (tree marker sync, visible-tree listing, search aggregation).